

Cross-Layer Anomaly Detection in an SDN-Enabled Heterogeneous Network

Team 14

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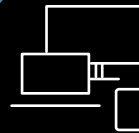
Problem and Motivation

Problem Statement: Design and development of SDN-controlled heterogeneous network testbed for cross-layer telemetry-based anomaly detection.

- Alex a Network Operations Engineer is flooded with alarms from multiple dashboards. Customers are complaining, services are down. Hours spent on investigation, yet nobody knows where the problem actually started.
- The real challenge is an anomaly that can propagate and appear as a performance problem.

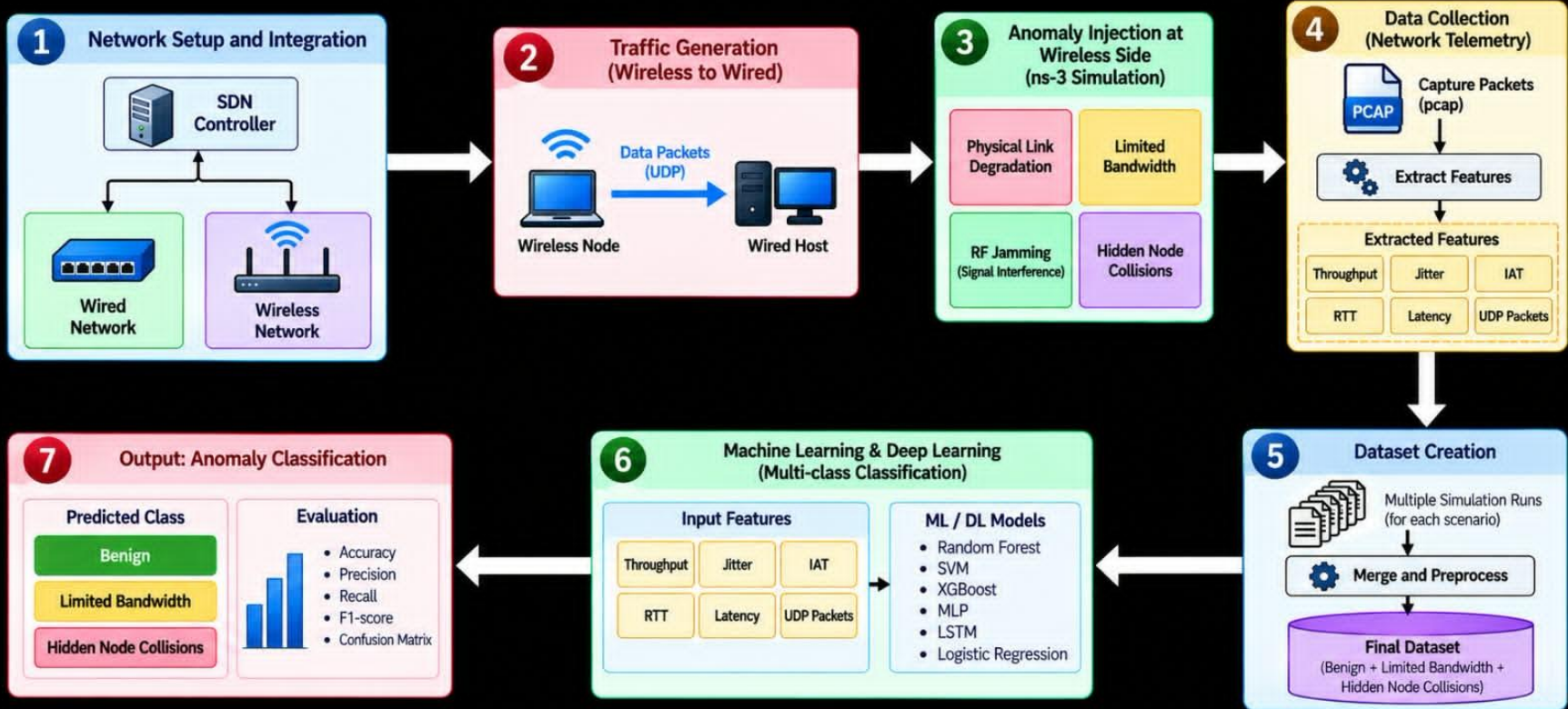


What if we could collect telemetry from multiple layers, correlate the information, and automatically detect anomalies under centralized SDN control?



Alex would be able to detect anomalies earlier, identify the root cause faster, and reduce troubleshooting effort and downtime.

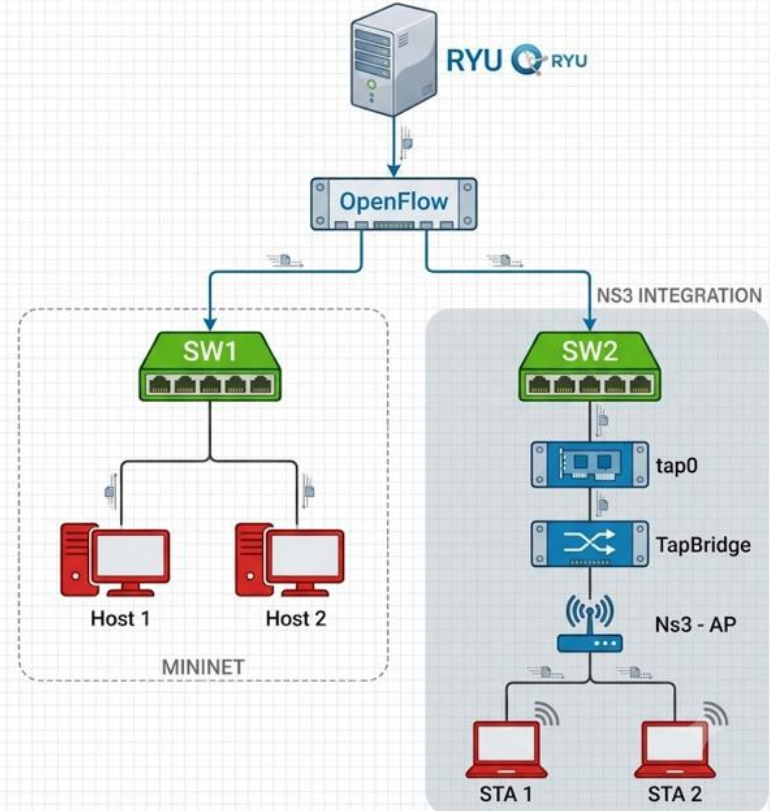
System Workflow



Tools Used & Topology

- **Mininet:** Emulates wired topology, hosting virtual endpoints (h1, h2) and OpenvSwitch instances (sw1, sw2)
- **Ns-3:** Simulates 802.11g wireless network (AP, STAs) & executes wireless fault injections.
- **Ryu-SDN Controller:** Centralized control plane executing Layer2 learning switch logic and managing forwarding state.
- **OpenFlow(v1.3):** Southbound protocol, installing dynamic flow match rules from Ryu down into switches.
- **Linux TAP & TapBridge:** Bridges Mininet OVS bridge to Wi-Fi stack via **tap0** in **UseBridge** mode, for cross layer telemetry collection.

HYBRID SDN NETWORK EMULATION-SIMULATION TOPOLOGY

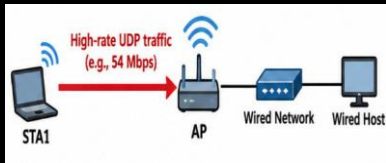


Fault Scenarios

Limited Bandwidth

Generates a continuous 54 Mbps UDP flood using large 1024-byte packets.

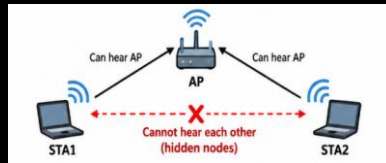
This Uplink DoS exhausts the 802.11g channel capacity, forcing the Access Point's transmission queues to overflow.



Hidden Node Collisions

STA 1 and STA 2 Nodes are physically separated so that their carrier sense mechanisms cannot detect each other.

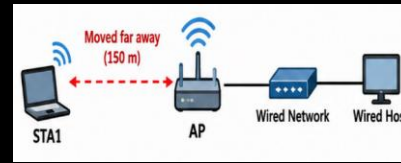
Simultaneous transmissions physically collide at the Access point antenna, destroying the packets.



Physical Link Degradation

Manipulates node coordinates to mid simulation to force a 150-meter separation.

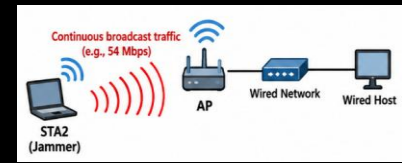
This abruptly drops the Signal-to-Noise Ratio (SNR), causing MAC frame drops and total loss of Wi-Fi association.



RF Jamming

A rogue node blasts continuous raw radio frequency energy across Wi-Fi Channel 1.

Legitimate nodes sense a busy medium via CSMA/CA and pause transmissions, causing extreme queue delays.



Methodology

01

EMULATE

STACK

Mininet 2.3 · OVS
OpenFlow 1.3 · NS-3

TOPOLOGY

2 switches · 4 hosts
1 AP · 2 stations

TRAFFIC

UDP traffic

RADIO

NS-3 802.11g
log-distance exp 3.5

02

INJECT

WIRELESS

Limited BW, Collision
RF Jamming

GRID

4 fault types
× 2 severity levels

SCHEDULE

Multiple 3000 episodes
randomised order

03

OBSERVE

OPENFLOW

port + flow counters

QUEUE

backlog · drops

RADIO

RTT · JITTER
retries

END-TO-END

UDP
ping

04

LEARN

FEATURES

counters → rates
5 stats per window

MODELS

LR · LSTM
SVM · RF · XGBoost

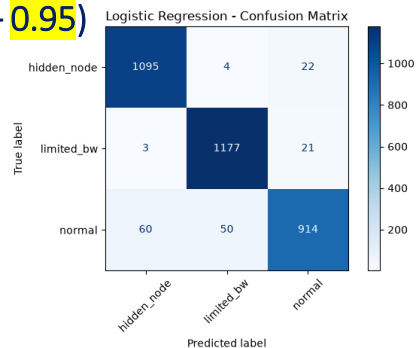
EVALUATION

Macro F1, Precision,
Recall, Accuracy

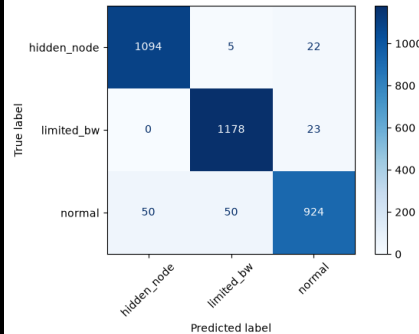
Results

Logistic Regression (Macro F1 – 0.95)

	Pre	Rec	F1
Normal	0.96	0.89	0.92
Limited Bandwidth	0.96	0.98	0.97
Hidden Node Collisions	0.95	0.98	0.96



SVM - Confusion Matrix

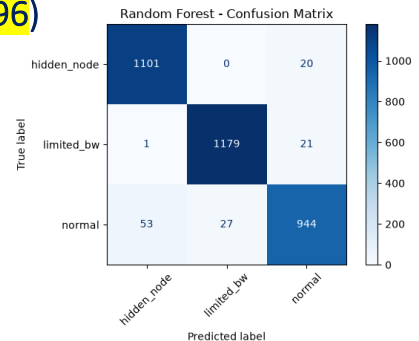


SVM (Macro F1 – 0.95)

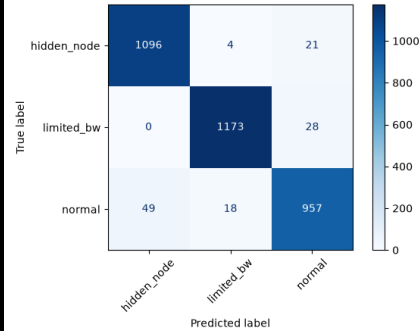
	Pre	Rec	F1
Normal	0.95	0.90	0.93
Limited Bandwidth	0.96	0.98	0.97
Hidden Node Collisions	0.96	0.98	0.97

Random Forest (Macro F1 – 0.96)

	Pre	Rec	F1
Normal	0.96	0.92	0.94
Limited Bandwidth	0.98	0.98	0.98
Hidden Node Collisions	0.95	0.98	0.97



XGBoost - Confusion Matrix



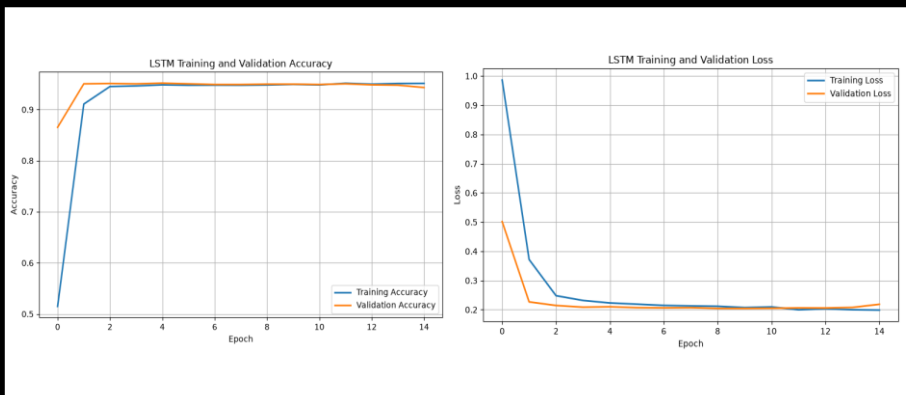
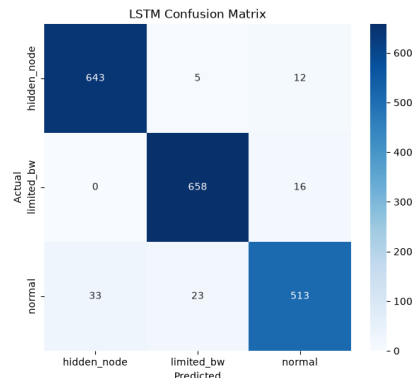
XG Boost (Macro F1 – 0.96)

	Pre	Rec	F1
Normal	0.95	0.93	0.94
Limited Bandwidth	0.98	0.98	0.98
Hidden Node Collisions	0.96	0.98	0.97

Results

LSTM (Macro F1 – 0.95)

	Pre	Rec	F1
Normal	0.95	0.90	0.92
Limited Bandwidth	0.96	0.98	0.97
Hidden Node Collisions	0.95	0.97	0.96



Q: Why the Ensemble Models Outperformed LSTM?

Better suited for tabular data:

Ensemble Models effectively learn patterns directly from structured network features such as latency, jitter, packet loss, and throughput.

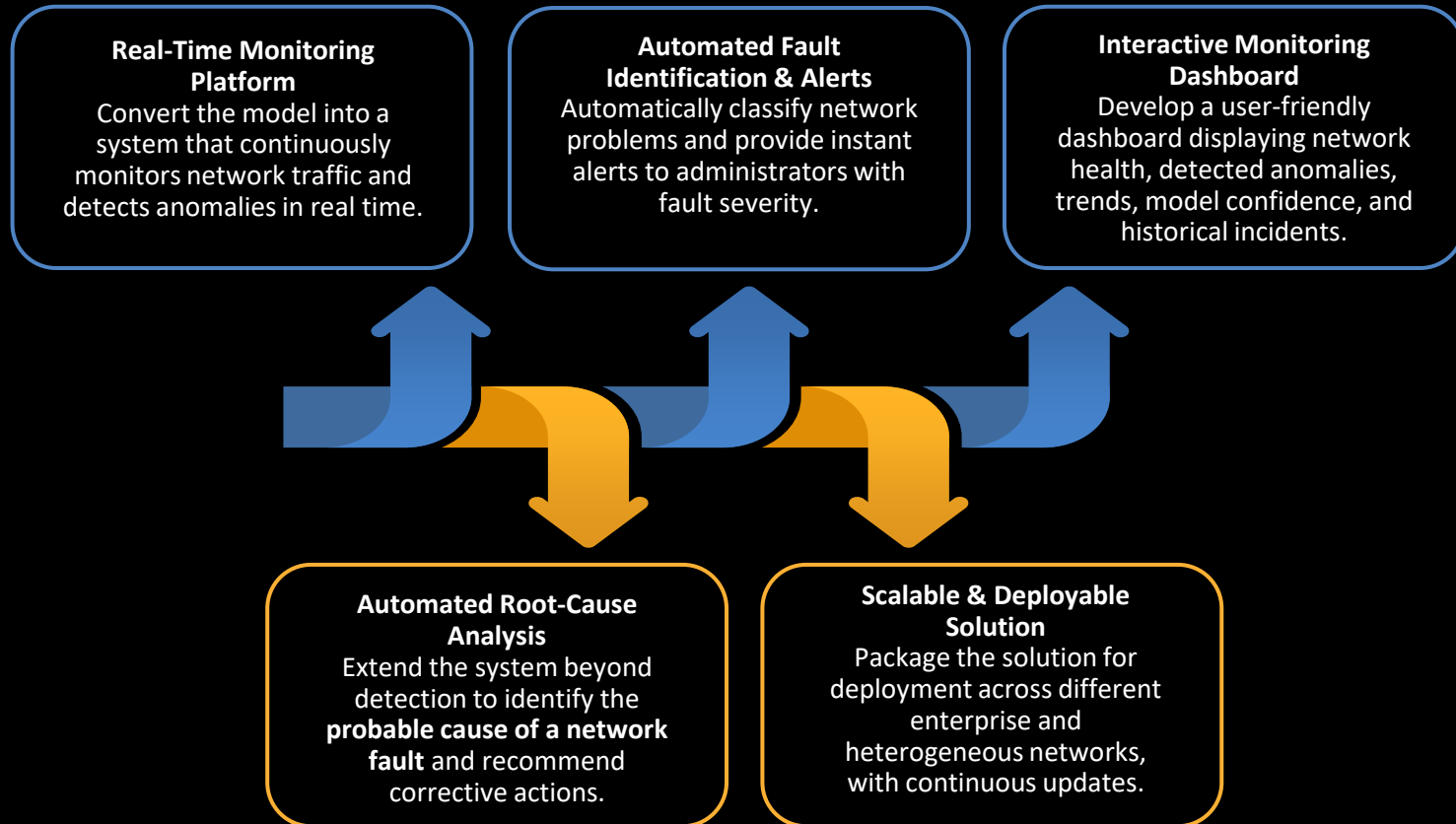


Limited temporal dependency:

Although the dataset contains temporal sequences, the fault classes appear to be more strongly distinguishable from feature-level patterns than from long-term temporal dependencies.



Future Scope



Thank You

